SPRINT 2

# Text Extraction

Skill extraction & gold-set evaluation

**Group 3** · October 9, 2026

Taxonomy of Work Capstone · Northeastern × Workhuman

# WHAT WE'RE BUILDING

Raw public job postings → organized, benchmarked skill taxonomy

## 1 Collect

Gather public postings from approved sources

Sprint 1 · done

## 2 Extract

One atomic skill statement per skill, and measure its quality

This sprint

## 3 Organize

Cluster, label, build the hierarchy, benchmark vs ESCO / O*NET

Next sprints

# SPRINT 2 GOALS

## 01

### Choose the approach

Pick an extraction method  and be able to justify why.

## 02

### Build a gold set

Hand-label 100 postings as a trusted reference to measure against.

## 03

### Measure honestly

Report precision / recall, and log every LLM token used.

# ONE CORPUS, CLEANED — NOTHING DELETED

| Corpus | Count |
|---|---|
| Postings | 123,849 |
| Exact duplicate descriptions (tagged) | 16,320 |
| Too short (<200 chars) / missing | 636 / 7 |
| **Usable unique postings** | **106,909** |

### Tag, don't drop

Every row stays, flagged: is_duplicate, is_short, is_usable.

### Line breaks kept

Bullets stay readable for labelers and the LLM.

### Contacts masked

Emails and phone numbers removed before sharing.

*Source: Kaggle LinkedIn Job Postings (arshkon), Mar–Apr 2024 · CC0 license*

# GOLD SET: 100 POSTINGS, EVERY SECTOR

| Sector | Postings |
|---|---|
| IT | 15 |
| Health Care | 15 |
| Engineering | 12 |
| Finance + Accounting | 12 |
| Sales + Marketing | 12 |
| Management | 10 |
| Legal | 8 |
| Admin / HR / Customer Service | 8 |
| Mixed | 8 |

**100**

postings sampled

**1,317**

gold skills labeled

**seed 42**

reproducible sample

**1 → 0**

posting with 0 skills (kept)

*Stratified by LinkedIn's first-listed job function · 300–5,000 chars · usable rows only. Noisy tags corrected with sector_fixed (an 'Attorney' was tagged IT).*

# WHAT COUNTS AS A SKILL

Clear rules before labeling · data/gold/LABELING_RULES.md

## ✓ Label it

**Tools:** Excel, Salesforce, Epic, Westlaw

**Methods:** phlebotomy, financial modeling

**Knowledge:** GAAP, HIPAA compliance

**Soft skills:** written communication

**Languages:** Spanish

## × Do not label it

**Degrees & licenses:** certifications

**Experience:** years of experience

**Traits:** 'self-starter'

**Boilerplate:** benefits, company info, EEO

**Vague duties:** 'manage the team'

**Duty rule:** a duty that names a skill counts — 'build prototypes' → prototyping. One skill per item, posting wording, no filler ('strong', 'proficiency in').

# HOW WE LABELED: LLM-SUGGESTED, HUMAN-DECIDED

**LLM suggests skills** → **Human accepts / edits / rejects** → **Both lists saved → edits measurable**

## Validation of doubtful skills

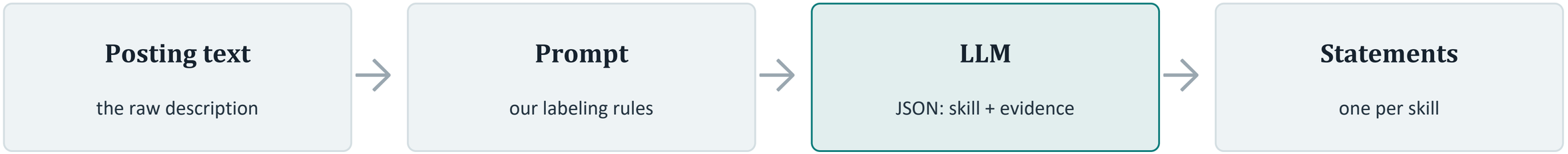
- **ESCO v1.2.1** (13,939 skills)
- **O*NET 31.0** (31,821 tool rows)
- Looked up via our check_skill.py, plus the web for new tools

**Finding:** ESCO misses many US legal procedures (depositions, written discovery) and niche tools — 'not in ESCO' does not mean 'not a skill'.

*Reported honestly: 3 postings double-labeled (exact-name agreement 30–52%, mostly wording) · 12 rows kept the AI suggestion unchanged.*

# EXTRACTION APPROACH

LLM prompting with structured output




*Reliability layer: cache on disk · rate-limit retries · token log*

## Why an LLM

Skills appear in free text, duties and lists, with many wordings — rules and NER miss most of them.

## Model

Qwen 3 (27B) on the Groq free tier. Temperature 0, JSON output. Total cost $0.

## Independence

Avoided GPT / Claude as the extractor — they suggested the gold labels, so reusing them would grade their own work.

*Gemini Flash was also tried, but hit 'high demand' (503) errors — only 7 / 100 finished.*

# Results: F1 = 0.72

**0.72**

**Lenient F1**

100 hand-labeled postings

*Strict F1 = 0.63*

| Matching | Precision | Recall | F1 |
|---|---|---|---|
| Strict (same name) | 0.60 | 0.66 | 0.63 |
| **Lenient (word overlap)** | 0.69 | 0.76 | **0.72** |

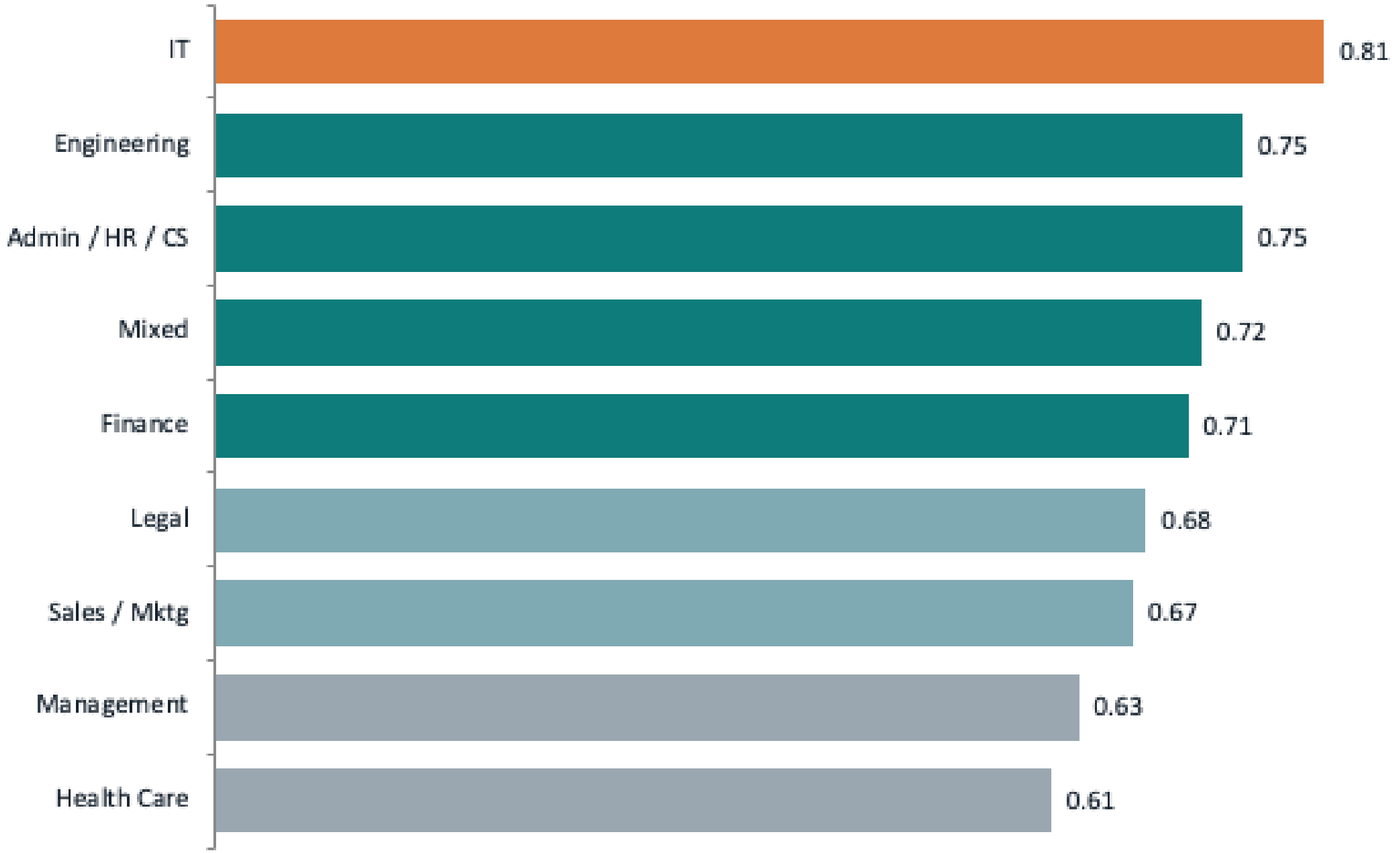
| LLM tokens | Input | Output | Total |
|---|---|---|---|
| All 100 postings | 91,554 | 41,902 | 133,456 |
| Per posting (avg) | 916 | 419 | 1,335 |
| Per extracted skill | — | — | **~92** |

✓ **100% of evidence copied verbatim — no invented text**

✓ **Zero-skill posting → model correctly returned 0**

# Results by sector

Lenient F1, by corrected sector




### Best — IT (0.81)

Tools have clear, unambiguous names — SQL, Snowflake, Jira.

### Weakest — Health Care, Mgmt

Duties and skills blur — 'patient care', 'safety checks'.

*Sector counts use sector_fixed, so they differ slightly from the gold-set slide.*

# ERROR ANALYSIS

**320 missed gold skills**

**451 extra predicted skills**

| Error type | Example |
|---|---|
| **Granularity (split vs merged)** | gold 'SAP/Chempax/Concur' merged → model lists SAP, Concur separately |
| **Wording** | 'preparing discovery' vs gold 'drafting discovery' |
| **Duty vs skill boundary** | model adds 'patient care', 'safety checks', 'requirements gathering' |
| **Soft skills / traits** | model skips 'organized', 'multitasking'; humans kept them |
| **Gold inconsistency** | some gold rows rename or merge skills against our own rules |

**Takeaway:** many 'errors' are gold-set wording, not model mistakes → next step is to normalize the gold names.

# LLM USAGE: FREE TIER, FULLY LOGGED

| Metric | Value |
|---|---|
| Provider / model | Groq — qwen/qwen3-27b |
| API calls | 100 (paid once, then cached) |
| Input tokens | 91,554 |
| Output tokens | 41,902 |
| **Cost** | **$0 (free tier)** |

**$0**

total cost

**100**

calls (cached)

**133K**

tokens

**Gemini attempt:** 7 postings before 503 capacity errors. **Claude Code** was used to build and test the pipeline; usage is documented in the sprint report.

# Limitations & next steps

## Limitations

- Gold labels were LLM-suggested; 12 rows unchanged → possible anchoring
- Only 3 postings double-labeled → weak agreement estimate
- Kaggle data is 2024, one LinkedIn-derived source
- Name-based matching (gold set has no evidence offsets)

## Next — Sprint 3

- Run the extractor on a stratified sample of the corpus → statements table
- Normalize gold names; adjudicate double-labeled postings
- Prompt-tune only on a dev split (avoid overfitting the 100)
- Embed and cluster statements; map to ESCO / O*NET

# Thank you

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